

AI Assistant

BTFViewer is an **AI-assistant tool for RTOS trace analysis**: find evidence and explain. This file is the technical and user-facing reference for the **AI** tab — investigation, model configuration, GUI tools, workflows, evaluation, and implementation.

For day-to-day panel usage and the product-level AI overview, see [README.md](#) → [AI Assistant](#). For repeatable ask-order playbooks, see [WORKFLOWS.md](#).

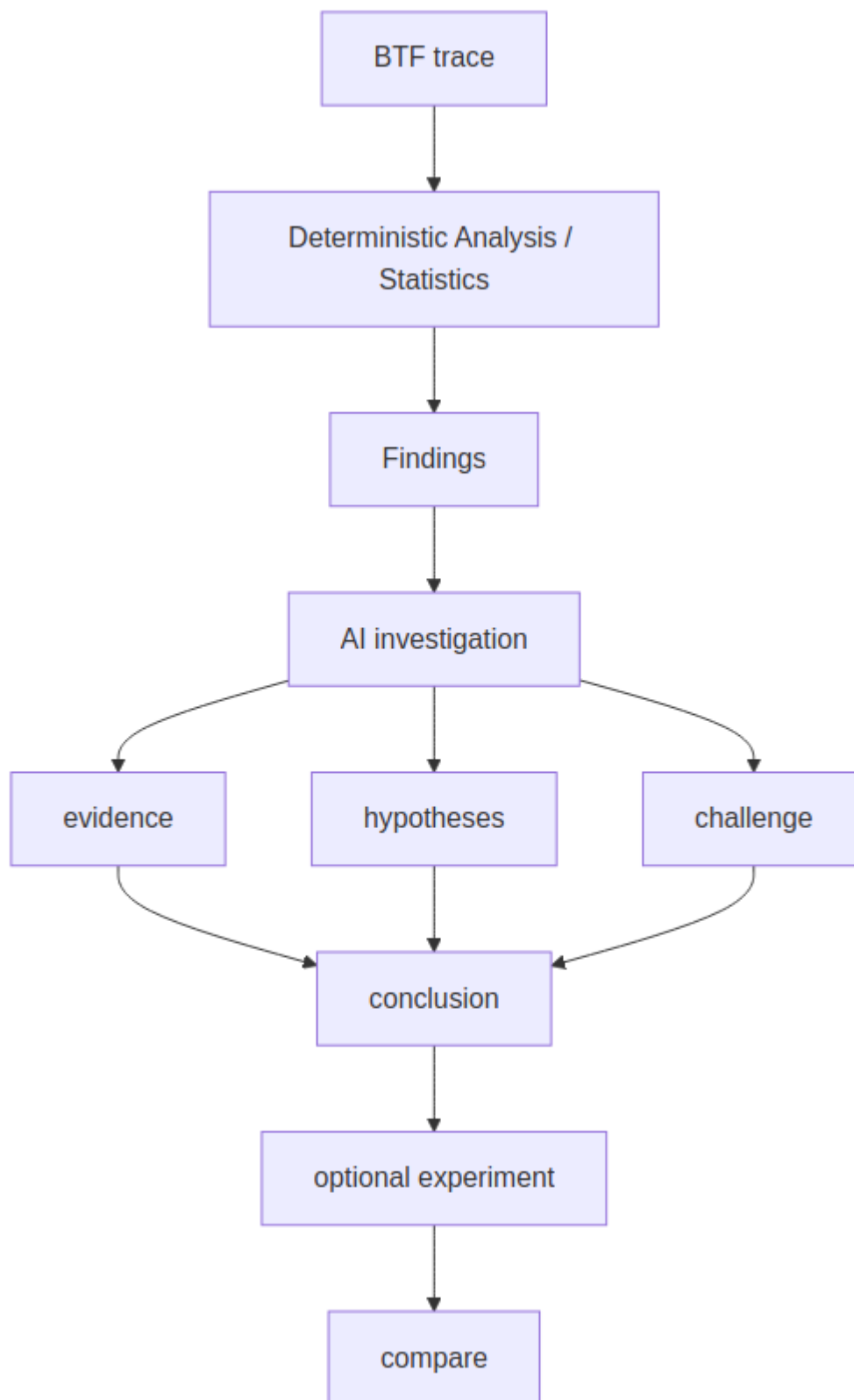
Core boundary: the AI works over structured BTFViewer findings, statistics, timeline queries, and comparison results. It does not read firmware source/ELF files, and what_if is a labelled heuristic slice-replay estimate rather than a FreeRTOS kernel simulation.

Documentation map

AI.md is the **AI system reference**, not the primary step-by-step troubleshooting guide.

Document	Responsibility
README.md	Product and user-facing AI entry: find evidence and explain
STATISTICS.md	Deterministic measurements AI consumes
WORKFLOWS.md	Practical investigation sequences and ask order
AI.md	Models, tools, planner, evidence validation, evaluation, implementation

AI boundary



AI may organize, explain, correlate, challenge, and estimate. It must not present heuristic estimates as measured trace results.

Documentation boundary

- **README.md** — user-facing AI entry point and product behavior
- **WORKFLOWS.md** — repeatable diagnosis and AI ask order
- **STATISTICS.md** — deterministic measurements consumed by AI
- **AI.md** — models, tools, planner, evidence validation, evaluation, and implementation

Contents

#	Section	Purpose
1	How it works	Context, evidence, tools, validation
2	Workflows and use cases	End-to-end investigation patterns
3	Endpoints and models	Providers, model capability, privacy
4	GUI tools	Tool groups, beginner mental model, Apply / Undo, complete schema
5	Desktop vs web	Platform differences
6	Troubleshooting	Common configuration/runtime problems
7	Opening the web app from file://	Local web + Ollama CORS
8	CLI regression gate	Headless analysis and AI tests
9	Benchmark / evaluation suite	Dataset, models, metrics, reproducibility
10	Investigation Case	Case/evidence model
11	Investigation planner	Planning and evidence sufficiency
12	Causal and temporal engines	Host-side heuristic reasoning
13	Implementation notes	Shared engines, validator, parity
14	Diagrams	Mermaid / response visualization

How it works

The **AI** tab answers diagnostic questions using structured **Analysis Findings** and summary metrics—not the raw .btf event stream. That keeps the prompt compact (token-efficient) and analysis fast. Toolbar **Compare** (right after **Analysis**, when two or more tabs are open) diffs the traces; the **Trace Compare** template / **Query with AI...** in that dialog sends those tables instead of Findings. Compare **Trends** lists every open tab; **Save as baseline** / **Score vs baseline** use the same stored profile as `baseline_score` (not a new tool). Analysis Findings **Save recipe...** / **Story...** are host chrome (user template / story export). **Ctrl+K** jumps to Analysis, AI, Compare, workspace presets, and Inspect task.

When a question needs per-task time-series, Find hits, Trace Compare, or a root-cause chain, the model calls tools from **GUI tools** — it still never reads the raw .btf file. Read-only batches run immediately; mixed GUI batches wait for **Apply** unless **Auto-apply GUI actions** is on. Export tools still show a save dialog.

Important conclusions should cite evidence (`jump:TIME`, metric names) and state **confidence** (High / Medium / Low) plus evidence quality (Directly observed / Strong correlation / Possible explanation / Insufficient evidence). The system prompt asks the model not to invent numbers, task names, or `jump:TIME` values absent from findings, tool results, or Trace Compare tables — and, when a **Cursor region window** is listed, to cite only times inside that window.

The AI panel stepper (Triage → Scope → Investigate → Verify → Experiment → Compare) follows Findings and tools already run; click a completed stage to jump to that output in the log (the matching message flashes). Empty log: **Start Investigation** runs **Auto investigate**. **Clear** removes chat replies, resets the usage meter, and clears current investigation issues (the stepper returns toward idle / **Start Investigation**). The investigation case, plan, and recent chat restore after app restart **when the log still has a user or assistant turn** (desktop [ai] `investigation_session`; web session snapshot). An empty log (or Clear then quit) does not restore a Current Issue card — **Start Investigation** stays available. What-if / Optimize stay on **Verify** until verify tools or strong evidence quality; the Experiment banner is a heuristic estimate (recapture a trace and Compare to measure). **Investigate** / **Root cause** / **Verify finding** / **Auto investigate** / **What-if** / **Optimize** /

Diagnostic report show an **Investigation plan** checklist (steps advance as tools run; the final text reply marks remaining steps done). Analysis Findings may include anomaly rows (WCET Max>>Avg spikes, extreme migration bursts). **Explain finding** (Analysis **Explain...**: Quick / Technical / Deep) calls `explain_finding` with `level=`. Mode chips and **More → Investigations** (including **Save as template...**) start tool sequences without adding templates after Auto investigate. Diagnose / Investigate / Auto investigate walk graph → temporal → rank_root_causes → challenge_conclusion before what_if. Evidence hypothesis rows offer Support / Reject / Need evidence / Test / Compare.

Right-click a timeline segment → **Ask AI about this event** to ask about that exact task/segment (Explain the timeline event for task ... around jump:TIME), same as Explain region but scoped to one segment. With **≥2 cursors**, the timeline context menu also offers **Explain this region with AI** (`explain_region`); the AI panel **Explain region** template is always available — without two cursors it runs on full-trace Findings (no region window). When **Settings → AI** is off, those timeline items are grayed (they open Settings if clicked). When cursors are placed, the prompt includes an explicit **Cursor region window** (`jump:lo ... jump:hi`) so replies should stay in-window. When investigate returns a root-cause chain and hypotheses, the Evidence panel renders a small **Investigation tree** plus an **Evidence Quality** meter (Strong / Medium-High / Medium / Weak — a diagnostic heuristic, **not** a probability). Direct evidence times, timeline correlation, and metric checks raise the band; untested alternatives lower it. The panel also lists **what would disprove** the conclusion, **evidence coverage**, and an **evidence graph** (finding → evidence → hypotheses). After the final reply, a host-side **validator** flags invented task names and `jump:TIME` values outside the cursor window.

Natural-language timeline questions (e.g. “find STI wait around TaskA”) are answered via `search_timeline` — no separate search UI.

Recommended ask order ([WORKFLOWS.md §7](#)): triage overall findings → **Investigate / Root cause** when you want tool-driven drill-down → named metric templates → mitigations after the timeline agrees. Prefer the built-in templates; they already name the metrics and units, and they point at Statistics pages that already exist (**Timeline Anomalies, Worst Events, Response Time, Critical Path, Period / Jitter, Unified Jitter, Recurring Patterns, Task Health, Task × Core, Core Utilization Over Time, Preemption Matrix, Waiter × Owner, Mutex Blocking**) — there is no extra `detect_timeline_anomalies` tool. Concrete flows and use cases are below.

Workflows and use cases

For the practical symptom-to-metric ladder and repeatable investigation sequences, use **WORKFLOWS.md**. This section documents the AI-side behavior and capabilities behind those workflows.

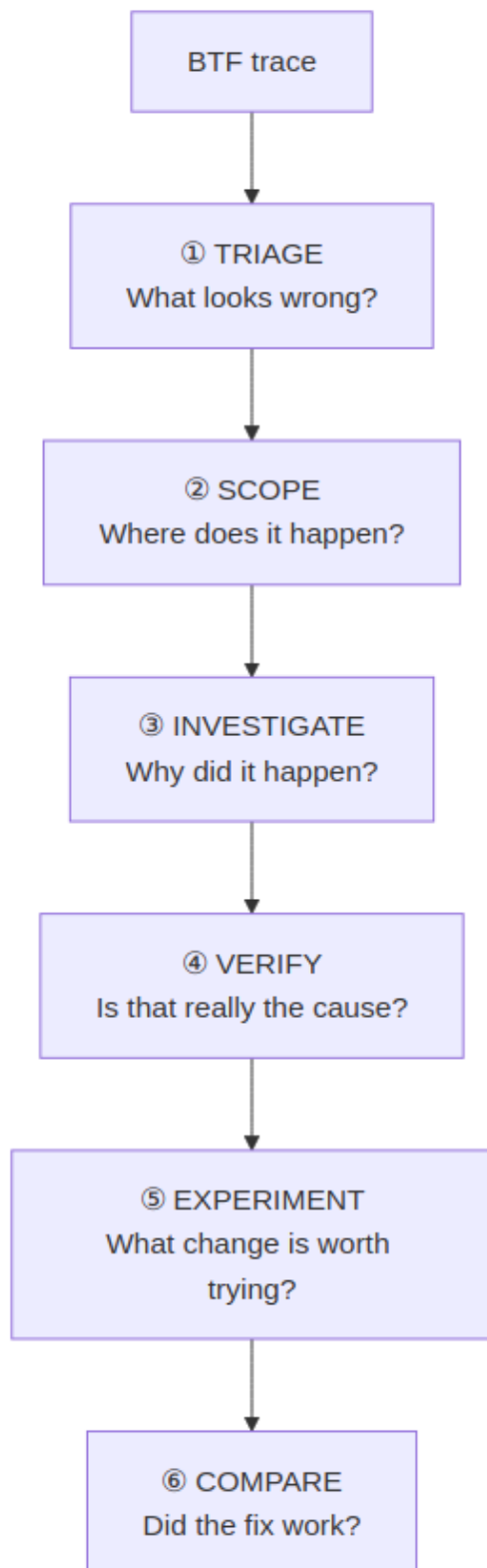
Same flow on **Desktop** and **Web**. Panel chrome: [README](#) → [AI Assistant](#). Ladder and ask-order tables: [WORKFLOWS.md §7](#).

In this section

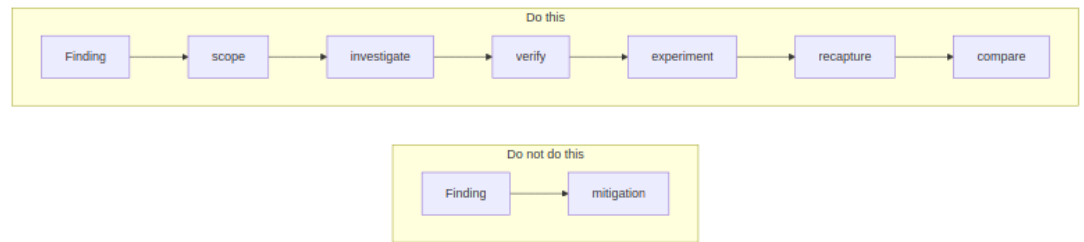
Newbie AI workflow	Triage → Scope → Investigate → Verify → Experiment → Compare
End-to-end flow	Full AI path from Findings to report
Investigation workflow	Triage → Investigate / Auto / Verify → evidence
Explain region and Ask event	C1-Cn window and segment context menus
What-if and optimize	Heuristic slice-replay (not FreeRTOS)
Use cases	Symptom → template / tools
Worked examples	Thrash, contention, regression, region, PI
Simulator limits	What <code>what_if</code> does and does not do

Newbie AI workflow — “I don’t know what is wrong”

For a first investigation, do **not** start by choosing individual tools. Use the six-step loop below and let **Investigate** choose the deeper tools.



Step	Start with	What you should get	What you must verify
1. Triage	Triage findings or toolbar Analysis	Ranked Critical / Warning / Info issues and a suggested place to start	The named Statistics page contains the same problem
2. Scope	Select the finding; apply suggested cursors or place C1-Cn	One task / incident / time window instead of the whole trace	Limit to C1-Cn is on when the question is phase-specific
3. Investigate	Investigate ; use Root cause when a suspect task is already known	Hypotheses, related evidence, critical path / dependency / temporal relationships	Click <code>jump:TIME</code> , <code>range:L0/HI</code> , and named Statistics sections
4. Verify	Verify with AI... or continue the Investigation plan	Supported / rejected / insufficient conclusion, alternatives, contradictions, evidence quality	Evidence is in scope; task names and times are real; alternatives were challenged
5. Experiment	What-if, Optimize , or experiment plan	Ranked <i>estimated</i> pin / priority / contention / migration changes	Treat the result as an estimate only; change firmware/configuration and recapture
6. Compare	Open before/after tabs → Compare → Query with AI... / Validate experiment...	Measured deltas and whether the expected improvement happened	Same workload phase and comparable cursor scope on both traces



The beginner rule

A useful first session therefore needs only these user-facing actions:

If you want to...	Use
Find the biggest issue	Triage findings
Understand one issue	Investigate
Check whether the explanation is real	Verify with AI...
Understand one selected time range	Explain region
Test a possible change	What-if / Optimize
Prove the change helped	Trace Compare

Individual tool names such as `correlate_events`, `find_critical_path`, or `rank_root_causes` are mainly useful for advanced users and developers. The normal workflow should be driven by the templates and Investigation plan.

End-to-end flow



Do not ask for mitigations before the timeline agrees with the finding. Empty or mis-scoped Statistics produce confident nonsense. Prefer built-in templates: they already name metrics and units.

Investigation workflow

Step	Template or tool	Why
1	Triage findings / detect_anomalies	Rank Critical / Warning / Info; open Timeline Anomalies / Worst Events / Task Health
2	Investigate / investigate — or Findings Auto investigate...	Root-cause chain, hypotheses, alternatives, suggested tools

Step	Template or tool	Why
3	Verify finding / Findings Verify with AI...	Confirmed / Rejected / Inconclusive with jump:TIME evidence
4	correlate_events + query_raw_metric	Merge blocking / execution / migrations / sync / PI for one task
5	find_critical_path / detect_priority_inversion	Preempt/block path; L/M/H inversion suspects
6	build_task_dependency_graph / analyze_temporal_causality	BTF wait/preempt/migrate chain
7	rank_root_causes / challenge_conclusion	Rank then alternatives before what_if
8	find_related_findings / compare_tasks	Adjacent findings; side-by-side task deltas
9	set_cursors / zoom_to_range / highlight_task / bookmark_finding	Narrow the timeline (Apply cursors unless auto-apply is on); click range:L0/HI / btfrange: on critical-path evidence
10	Evidence panel	Investigation tree + Evidence Quality + what would disprove this
11	investigation_replay / generate_report / export_investigation	Structured close-out; optional export_report

Root cause walks deadline/WCET → preemption → blocking → mutex → inheritance → migration for the top finding. Use it when triage already named a suspect task.

Auto investigate chains verify-style steps for one finding (investigate → correlate → critical path / graph / temporal → rank → challenge → what-if) and advances the Investigation plan checklist. Use **Verify** when you already have a finding id and want a short verdict.

Explain region and Ask event

Entry	When it appears	Scope
Timeline → Explain this region with AI	Only with ≥2 cursors ; grayed when AI is off	C1-Cn; prompt gets Cursor region window: jump:lo ... jump:hi
AI panel → Explain region	Always	Same window when ≥2 cursors ; full-trace Findings if none
Timeline segment → Ask AI about this event	Segment under the pointer; grayed when AI is off	One task / core / segment around jump:TIME

Stay inside the stated window: every jump:TIME in the reply should fall between C1 and Cn (or the model should say the window has no matching evidence). Enable **Limit to C1-Cn** so Statistics / Findings / query_raw_metric match that window. Clicking a jump:TIME outside the cursors usually means the model invented or reused a full-trace time — discard it and re-ask with cursors + scoped Findings.

What-if and optimize workflow

what_if and optimize_experiment are **heuristic slice-replay** tools: they reallocate measured execution slices, scale migrations / blocking, and adjust core-util balance. They are **not** a FreeRTOS kernel or deterministic scheduler. Every result carries a disclaimer. After a promising estimate, recommend_experiments suggests validation steps (simulation / firmware / measurement).

Goal	What to run	Typical change phrases
One concrete idea	What-if → what_if	pin CS[28] to Core_0, raise priority of Low[266], reduce mutex contention 50%
Rank several ideas	Optimize → optimize_experiment (then optional optimize for qualitative notes)	Host picks pin-to-dominant / quiet core, contention –50%, priority up, migrations –50%
Soft advice only	optimize	Finding-text mitigations without scored experiments
What to try next on the bench	recommend_experiments	Validation experiments from findings heuristics

Reading the payload: compare baseline vs simulated (migrations, blocking_ns, load_balance_score) and the deltas.cost ranking. Lower cost is better in the experiment list. Treat Medium confidence as “worth trying on the bench”; Low means the phrase was vague or slices were thin.

Use cases

Situation	Before you ask	Template / tools	Then verify
Unknown — first look	Full-trace or cursor-scoped Findings	Triage findings → Investigate	Timeline Anomalies / Worst Events / Task Health; jump:TIME

Situation	Before you ask	Template / tools	Then verify
Hottest / noisiest task	Findings name a suspect	Task profile	Period / Jitter; Task Health; Task × Core; Execution / Blocking p95/p99
Tick jitter / missed ticks	Trace Health in scope	Tick health	Tick Distribution (not Period / Jitter — that page is task inter-arrival)
Confirm one finding	Select it in Analysis Findings	Verify with AI... / Verify finding	Evidence panel; timeline
Explain a time window	≥2 cursors; Limit to C1-Cn on	Context menu or Explain region	Only jump:TIME inside C1-Cn
One segment / ISR slice	Right-click the segment	Ask AI about this event	That task's row; nearby STI
Auto walk a finding	Select finding → Auto investigate...	auto_investigate	Investigation plan + Evidence
Migration thrash / ping-pong	Scope thrash window; Findings mention the task	Migration thrash → correlate_events → What-if pin / Optimize	Task × Core; Timeline Anomalies migration bursts; Migrations Rate/Ping; Heatmap / Chord; Core Affinity

Situation	Before you ask	Template / tools	Then verify
Priority inversion / PI boost	Inheritance finding or PI episodes in scope	Priority inversion / detect_priority_inversion → find_critical_path	Priority Inheritance; Mutex hold; Waiter × Owner (heuristic handoff)
High blocking / mutex wait	Scope the stall; suspect task known	Highest latency → query_raw_metric blocking/sync → What-if contention / priority	Worst Events; Waiter × Owner; Blocking p95/p99; Mutex hold; Priority Inheritance
WCET / deadline pressure	Thresholds set in Display → Analysis	WCET / hot CPU or Deadline / budget → check_budget	Timeline Anomalies / Worst Events; Period / Jitter; Task Health; Execution Max / p95 / p99; Deadlines
Compare two tasks	Both names known	compare_tasks	Execution / Blocking / Migrations side-by-side
Related findings	One finding selected	find_related_findings	Shared task / metric / nearby times

Situation	Before you ask	Template / tools	Then verify
Load imbalance across cores	Multi-core util in Findings	Core balance → analyze_traces (multi-tab) or What-if pin to quiet core	Task × Core; Load Balance Score; Concurrent Active; Core Time Breakdown
A vs B build regression	Two tabs open	Trace Compare (toolbar Compare → Query with AI...) / compare_performance / regression_explain	Compare summary strip; Trace Compare pages; same scope on both builds
Drift vs saved baseline	Baseline profile stored (rc / localStorage)	baseline_score	Flags z >2; re-capture if needed
Rank all open traces	≥2 loaded tabs	analyze_traces	Best tab vs Migrations / LB / missed ticks
Write-up for a review	Cause already confirmed	Diagnostic report → generate_report → export_report / export_investigation	Saved HTML/CSV/JSON; evidence times bookmarked

Situation	Before you ask	Template / tools	Then verify
CI gate vs baseline	Desktop CLI	[analyze] (#cli- regression- gate) --fail-on- regression (optional --ai)	Exit code + markdown narrative

Worked examples

Migration thrash → pin affinity

1. Place cursors on the thrash window; enable **Limit to C1-Cn**; re-open Analysis.
2. Run **Migration thrash** or **Investigate** until the hot task (e.g. CS[22]) and cores match the heatmap.
3. Ask **What-if**: *pin CS[22] to its dominant core* (or run **Optimize** for ranked candidates).
4. Read Δ migrations / Δ load_balance_score. If migrations drop but LB worsens sharply, try pin-to-quietest via optimize_experiment and compare ranks.
5. On firmware: set affinity / reduce bounce; re-capture a .btf and toolbar **Compare** the before/after tabs.

Mutex contention → shorter critical section

1. **Highest latency** / correlate_events for the waiter; confirm hold episodes in Mutex / Priority Inheritance.
2. **What-if**: *reduce mutex contention 50% for TASK* (or another %).
3. Expect lower blocking_ns in the simulated payload — still an estimate. Confirm by shortening the hold in code and re-tracing.

Two builds → regression narrative

1. Open baseline and candidate as tabs; match cursor scope if needed.
2. Toolbar **Compare**, then **Query with AI...** (**Trace Compare** template), or `compare_performance` then `regression_explain`.
3. Act only on High/Medium confidence deltas that Statistics on both tabs reproduce.
4. Optional: `optimize_experiment` on the candidate's hottest task to sketch mitigations (still heuristic).

Cursor window → Explain region

1. Place **C1** / **C2** on the phase of interest (e.g. 1.060 s ... 1.120 s); enable **Limit to C1-Cn**; re-open Analysis.
2. Right-click the timeline → **Explain this region with AI** (or AI panel → **Explain region**).
3. Confirm the user turn lists Cursor region window: `jump:lo ... jump:hi`. Reject any `jump:TIME` outside that window.
4. Follow up with `correlate_events` / `query_raw_metric` on tasks named in-window; bookmark evidence times.

Priority inversion finding → Verify

1. Open Analysis Findings; select a Priority Inheritance / inversion row.
2. **Verify with AI...** (or **Auto investigate...** for a longer chain).
3. `Expectdetect_priority_inversion/query_raw_metric(priority_inheritance)/find_critical_path`; read the Evidence score and investigation tree.
4. Click in-scope `jump:TIME` links; confirm L/M/H on the timeline and in Priority Inheritance statistics.

Simulator limits

Does	Does not
Replay measured slices / migrations / blocking gaps for the Statistics scope	Run FreeRTOS scheduling, ISRs, or cache models
Score pin / priority / contention / migration experiments	Guarantee WCET or deadline after a firmware change
Label every result as estimate / not measured	Replace timeline verification or a new capture

Phrase changes as **pin / affinity / priority / mutex / migration** so the simulator engages; vague text falls back to a qualitative estimate (simulator: none).

Endpoints and models

Any OpenAI-compatible endpoint works, including Ollama (<http://localhost:11434/v1>). Chat requests time out after 120s (**Stop** still cancels sooner). Give the endpoint at least an **8k** context window so a full Findings card plus a tool round still fits.

Local models: the shipped Ollama default is qwen3.5:9b. Pull with `ollama pull qwen3.5:9b`. Larger local ids (qwen3.5:27b, qwen3.8:27b, gemma4:26b) trade latency for capacity. 3B-class models often skip native tools, dump tool JSON as text, and fail the investigation suite — do not use them.

Import presets: `[examples/ai]` (`examples/ai/README.md`) ships `[ollama.json]` (`examples/ai/gemini.json`) (`examples/ai/gemini.json`), `[openai.json]` (`examples/ai/openai.json`), `[deepseek.json]` (`examples/ai/deepseek.json`), `[grok.json]` (`examples/ai/grok.json`), and `[presets.json]` (`examples/ai/presets.json`). Unknown preset names are added to the combo. Imported values fill **Settings → AI** (including checkbox flags when the file names them); review and confirm to save. Each preset keeps its own base URL, model, key, auth mode, and TLS flag.

Keys follow the same rules on Desktop and Web: Settings → AI → API key first, then OPENAI_API_KEY, then GEMINI_API_KEY, then OLLAMA_API_KEY. A local Ollama endpoint needs no key. Custom agents (Cursor, ...) have no extra env name in the GUI — paste the key on that preset. Live ai-test XML may use `<api-key env="VAR">`. Full examples: [README → API keys](#).

AI capability / model matrix

Capability	Small local	Local 9B+	Cloud
Basic Q&A	✓	✓	✓
Tool calling	△	✓	✓
Investigation (investigate / root-cause chain / hypotheses)	△	✓	✓
Complex reasoning (multi-step correlation, alternatives)	△	✓	✓
Large traces (big Findings card / long chat history)	△	△	✓
What-if / optimize (what_if, optimize_experiment)	✓	✓	✓

✓ reliable · △ inconsistent — works sometimes but often skips native tool calls, hallucinates numbers, or truncates on long context; always verify against the timeline before trusting a result.

Which model should I use?

If you...	Use
Want a local investigator (no key)	qwen3.5:9b — shipped Ollama default; best practical local on the investigation suite (~52s/case, 78 overall)

If you...	Use
Need more local quality	qwen3.8:27b (88, ~190s/case) or qwen3.5:27b (81, ~149s/case). gemma4:26b is slower than 9b and slightly worse (73, ~111s/case)
Have a large scope (many findings, long chat) or want the strongest reasoning	Cloud (gpt-4o, Gemini, DeepSeek, Grok) — mind the privacy trade-off
Handle confidential traces	Local Ollama regardless of size — nothing leaves the machine

Small local models often skip native tool calls and emit a fenced “btftool block instead — the viewer renders the same GUI cards either way (see **GUI tools**) — but investigation-heavy templates (Investigate / Root cause / Verify / Explain region / Auto investigate / Ask AI about this event / What-if / Optimize) need a tool-capable model such as qwen3.5:9b to reliably chain multiple calls.

Credential storage

	Desktop	Web
Where keys live	[ai] *_api_key in btf_viewer.rc next to the viewer	Browser localStorage (btf-viewer-settings-v1)
At rest	Encrypted as enc1:... (machine-bound; not portable to another host)	Plaintext in localStorage — treat as convenience only
Sent to the model	Never as a chat field; only as the HTTP Authorization / API header to the configured endpoint	Same
Clear	Settings → AI → clear key, or delete btf_viewer.rc AI keys	Settings → Reset / clear site data

What leaves the machine

What AI receives

- ✓ Analysis Findings (titles, severities, task names, heuristic text)
- ✓ Selected metrics / tool results the model requests
- ✓ Scoped timeline search hits and correlate windows
- ✓ Trace Compare CSV when that template runs
- ✓ User question + short chat history

What AI does NOT receive

- ✗ Raw .btf / .btf.gz file bytes
- ✗ The full unrequested event stream
- ✗ API keys inside the prompt body

	Local Ollama	Cloud endpoint
Trace file stays local	✓	✓
Findings / metrics leave the machine	No (loopback)	Yes — to that vendor
Raw BTF uploaded	No	No
API key required	Usually no	Usually yes

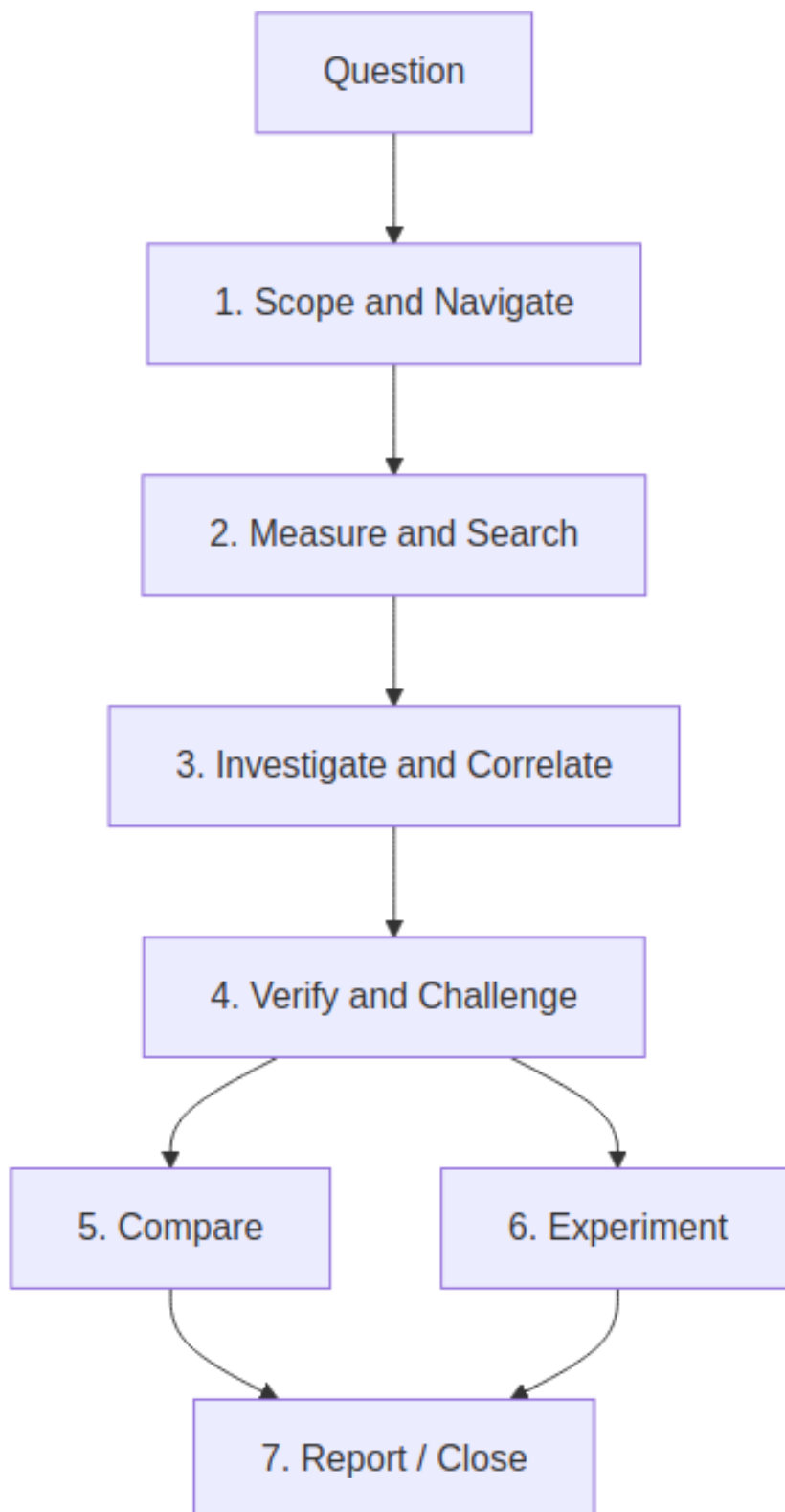
Prefer local Ollama for confidential traces. Redact sensitive task names in annotations before using a cloud preset.

GUI tools

Read-only evidence tools run immediately; GUI-changing tools wait for **Apply** unless **Auto-apply GUI actions** is on. Names and parameters are in [Complete GUI tool reference](#) below.

The AI tool set is easier to understand by **purpose** than by function name. A normal user should start with the built-in templates and Investigation plan; the individual tool schema is primarily an advanced/debugging reference.

Tool mental model



Apply, Skip, and Undo

Tools fall into two behavioral classes:

Class	Behavior	Examples
Read-only evidence tools	Run immediately; they do not change the viewer	query_raw_metric, search_timeline, investigate, correlate_events, find_critical_path, verify_claim
GUI-changing tools	With Auto-apply GUI actions off (default), the batch waits for Apply / Skip	set_cursors, zoom_to_range, highlight_task, set_view_mode, add_annotation, bookmark_finding

Several tool calls may arrive in one model turn and are applied as one batch. **Undo last actions** restores zoom / view / highlight / inspector / marks; Ctrl/Cmd+Z also reverts cursors and marks. Export tools still open a save dialog.

1. Scope & navigate — “Where should I look?”

Use these tools to turn an answer into a visible timeline location.

Goal	Main tools	Result
Scope a suspected phase	set_cursors, zoom_to_range	Places C1-Cn and focuses the relevant interval
Focus a task	highlight_task	Lock-highlights the task on the timeline
Change perspective	set_view_mode	Task/Core and horizontal/vertical view
Inspect a migration corridor	open_corridor_inspector	Opens the Migration & Corridor Inspector

Goal	Main tools	Result
Preserve evidence	bookmark_finding, add_annotation	Adds semantic or free-text timeline marks
Clean up	clear_marks, reset_view	Clears investigation clutter or restores full-span view

Beginner usage: normally accept these as GUI cards produced by **Investigate**, **Verify**, or **Explain region** rather than calling them directly.

2. Measure & search — “What happened?”

These tools fetch deterministic evidence. They do not change the trace.

Question	Main tools	Evidence returned
Where did an event occur?	search_timeline	Matching task/STI/tag/interval/pointer/migration timestamps
What are this task’s measured values?	query_raw_metric	Scoped execution, blocking, migration, sync, PI, or findings rows
Is the distribution unusual?	analyze_distribution	p50-p99.9, standard deviation, CV, outlier rate
Is timing periodic or jittery?	analyze_periodicity	Expected vs observed period/jitter statistics
Does the task exceed a budget?	check_budget	WCET/response/deadline budget comparison

These tools are the evidence layer. They should be preferred over asking the model to guess a number.

3. Investigate & correlate — “What is related?”

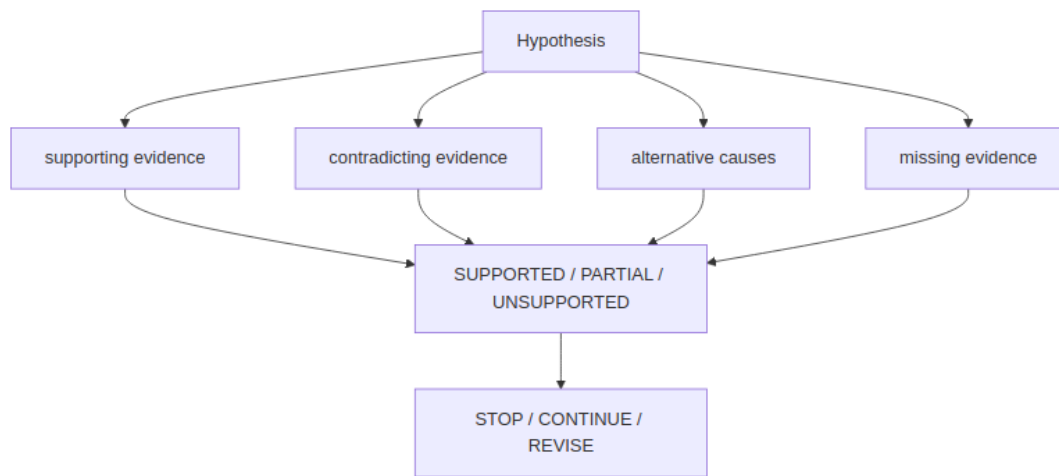
Use this group after Triage identifies a concrete issue.

Depth	Main tools	Purpose
Triage	<code>detect_anomalies</code> , <code>cluster_findings</code> , <code>cluster_incidents</code>	Rank and group Findings
Investigate	<code>investigate</code> , <code>plan_investigation</code> , <code>suggest_scope</code>	Build hypotheses and choose the cheapest next checks
Correlate	<code>correlate_events</code> , <code>find_related_findings</code>	Merge nearby execution/blocking/migration/sync/priority evidence
Path	<code>find_critical_path</code>	Walk preemption, blocking, and mutex activity around an incident
Dependency	<code>build_task_dependency_graph</code>	Show wait/preempt/migrate/PI relationships
Temporal	<code>analyze_temporal_causality</code>	Build a happens-before chain from observed evidence times
Root cause	<code>build_causal_chain</code> , <code>rank_root_causes</code>	Rank hypotheses while labeling causal/correlated/temporal relationships

Investigate / Root cause / Auto investigate orchestrate these deeper tools. Most users should not need to choose the sequence manually.

4. Verify & challenge — “Is this really the cause?”

This group prevents a plausible story from being treated as a confirmed diagnosis.



Purpose	Main tools
Check one claim	<code>verify_claim</code>
Find evidence against a hypothesis	<code>detect_contradictions</code>
Decide whether enough evidence has been gathered	<code>assess_evidence_sufficiency</code>
Force alternative explanations	<code>challenge_conclusion</code>
Track hypothesis state	<code>manage_hypotheses</code>
Inspect priority-inversion evidence	<code>detect_priority_inversion</code>
Evaluate investigation quality	<code>score_investigation</code>

The Evidence panel complements these tools with evidence quality, coverage, what would disprove the conclusion, and the investigation tree.

5. Compare — “What changed?”

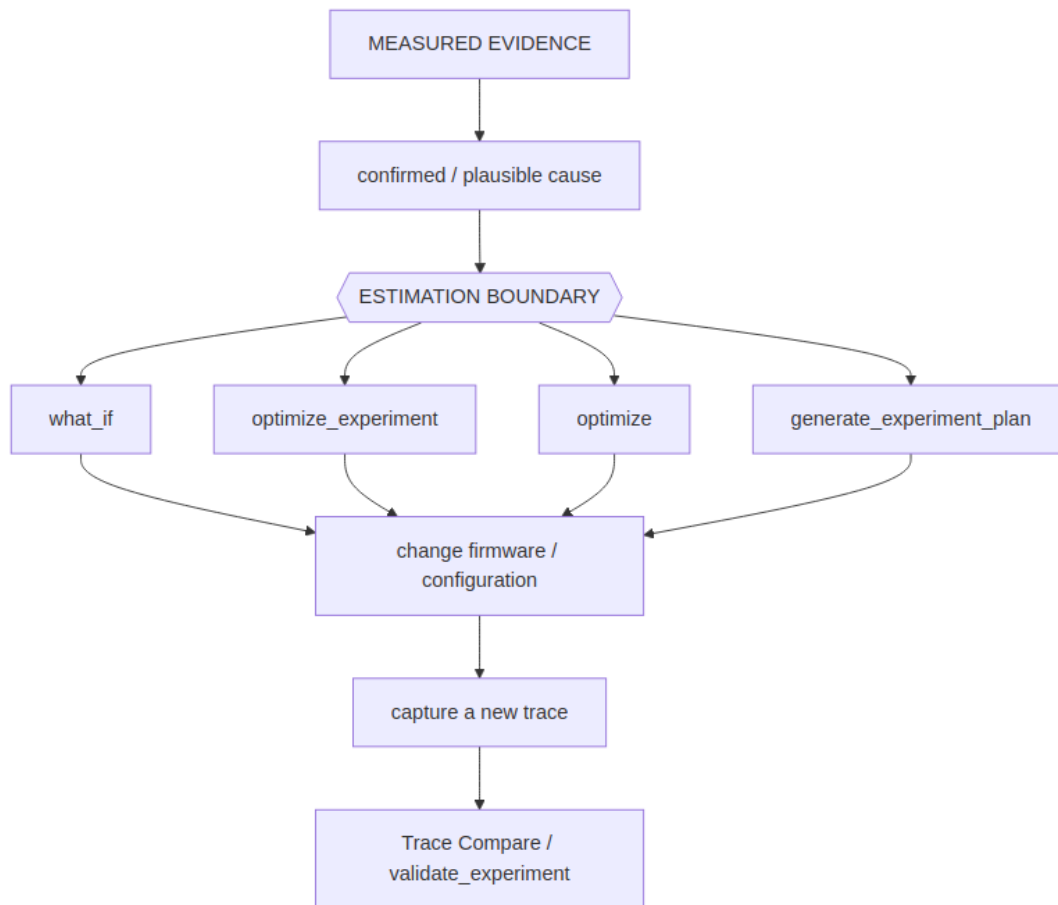
Use these tools when two or more traces are open.

Goal	Main tools	Result
Open / obtain Trace Compare	trigger_ compare	Same comparison data as toolbar Compare
Compare two builds	compare_ performance	Structured metric deltas + confidence
Explain the primary regression	regression_ explain	Narrative tied to A/B deltas
Localize the regression	regression_ localize	Suspect task / region / mechanism
Compare two tasks	compare_tasks	Side-by-side task metrics
Compare against history	baseline_ score	Drift against a stored baseline
Rank several open traces	analyze_ traces	Relative scheduling behavior

A comparison is only meaningful when the two traces represent equivalent workload phases.

6. Experiment — “What should I try?”

These tools operate **after** a cause has enough evidence.



Goal	Main tools
Test one concrete idea	<code>what_if</code>
Rank candidate changes	<code>optimize_experiment</code>
Get qualitative mitigation ideas	<code>optimize</code>
Generate bench/firmware validation steps	<code>recommend_experiments</code> , <code>generate_experiment_plan</code>
Compare prediction with measured result	<code>validate_experiment</code>
Store the outcome	<code>record_experiment_outcome</code>

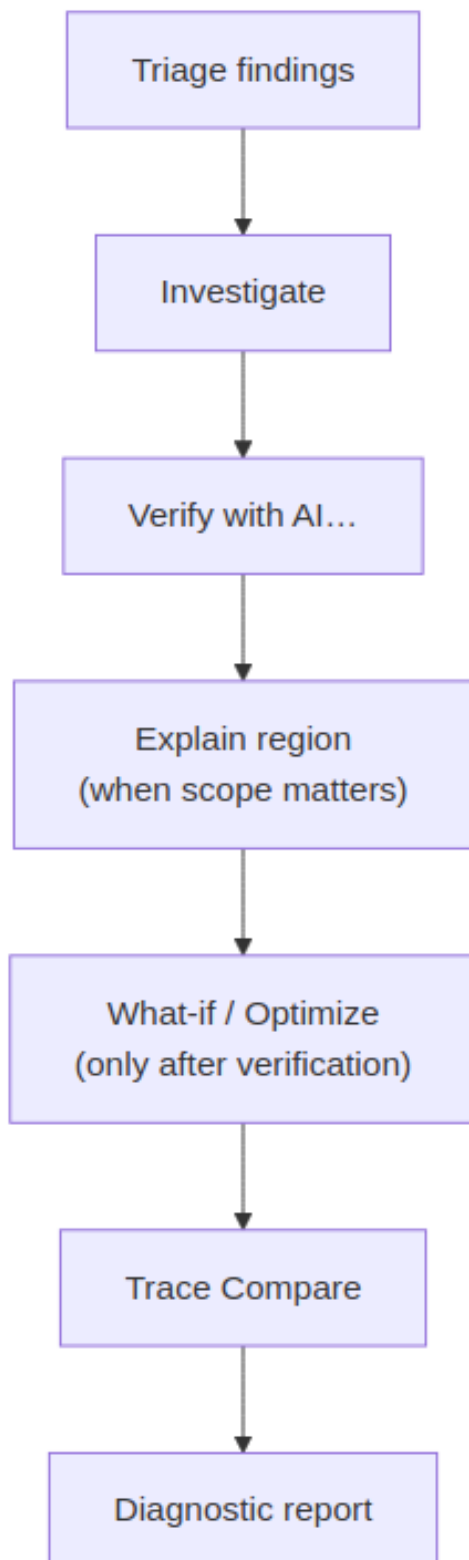
Important: `what_if` and optimization results are estimates, not measured scheduler behavior.

7. Report & close — “What did we learn?”

Goal	Main tools
Generate structured engineering text	<code>generate_report</code>
Save a report	<code>export_report</code>
Save a complete Investigation Case	<code>export_investigation</code>
Replay/summarize the investigation	<code>investigation_replay</code> , <code>summarize_investigation_context</code>
Remember similar cases	<code>investigation_memory</code> , <code>find_similar_investigations</code>
Finish the Case	<code>close_investigation</code>

Which tools should a beginner know?

Only a small subset needs to be visible in the mental model:



Everything else is supporting machinery selected by the planner or used by advanced users.

Complete GUI tool reference

The table below is the exhaustive schema reference. Use it when implementing, debugging, or explicitly steering tool calls.

Tool	Parameters / targets	Effect
set_cursors	timestamps (1-8 trace times)	Place cursors (enables Limit to C1-Cn when two or more)
zoom_to_range	start_time, end_time	Focus the timeline between two times
highlight_task	task_name_or_id (display name, numeric id, or merge key)	Lock-highlight a task row. Unknown names are ignored so the timeline is not dimmed. Empty string clears.
set_view_mode	mode (task / core); optional orientation	Switch Task or Core view; horizontal or vertical
open_corridor_inspector	optional core_from / core_to (Core_0, 0, c0, Core 0)	Open Migration Inspector; aliases resolve the same way
add_annotation	time, note (≤ 240 chars)	Pin an orange timeline note at a timestamp (stays on the current right-panel tab)

Tool	Parameters / targets	Effect
query_raw_metric	task, metric (priority_ inheritance, execution, migrations, blocking, sync, findings)	Read-only: return the per-task series for the current Statistics scope (up to 40 rows)
export_report	optional format (html / csv / json)	Download HTML/CSV/JSON bundling Analysis Findings, mermaid diagrams from the chat, annotations, and GUI state (cursors / highlight / view); json saves a full investigation package (see export_ investigation).
clear_marks	optional what (annotations / cursors / bookmarks / all / everything)	Clear AI clutter. all (default) drops annotations + cursors; everything also clears bookmarks
reset_view	(none)	Fit the timeline to the full span and clear the task highlight (marks stay)
search_timeline	query; optional mode (contains / exact / regex / sti / tags / intervals / lifecycle / pointers / migrations)	Find-panel search; returns matching timestamps (up to 40)

Tool	Parameters / targets	Effect
trigger_compare	optional tab_a / tab_b (0-based tab index or filename)	Read-only Trace Compare CSV + open the same dialog as toolbar Compare (needs two loaded tabs)
investigate	optional finding_id, depth (1-5)	Read-only: investigation graph with root-cause chain, hypotheses, ranked anomalies, suggested tools
detect_anomalies	optional limit (1-40)	Read-only: rank Analysis Findings as Critical / Warning / Info
correlate_events	task; optional around_time, window	Read-only: merge blocking / execution / migration / sync / priority / Find hits into one timeline

Tool	Parameters / targets	Effect
<code>find_critical_path</code>	task; optional timestamp, window (default 2000)	Read-only: preempt/block/mutex critical path around a timestamp; also returns a mermaid graph (graph LR), graph_nodes (id/label/kind/time), and split blocking_ steps / preemption_ steps arrays. Path steps carry start/stop; Evidence bullets use clickable range: L0/HI to zoom and place C1-C2 on the episode
<code>compare_performance</code>	optional tab_a / tab_b	Read-only: structured A vs B metric deltas + confidence (two tabs); data.regression_ type classifies the primary delta as execution / scheduling / synchronization / migration / load_ balance / unknown (legacy classification values like thrashing / load_imbalance / tick_health are preserved alongside it)

Tool	Parameters / targets	Effect
generate_report	optional report_type, finding_id	Read-only: typed engineering markdown (executive / performance / root_cause / regression / optimization / bug / ci); call export_report to save
check_budget	optional budgets, tasks	Read-only: compare per-task WCET/response/deadline metrics against budgets (host builds rows from findings when tasks omitted)
optimize	optional limit (default 5)	Read-only: evidence-backed mitigation ideas (estimate disclaimer)
regression_explain	optional tab_a / tab_b	Read-only: compare two tabs then narrate the primary regression; includes the same regression_type classification
bookmark_finding	time, kind (root_cause / evidence / correlated / reference); optional note	GUI: pin a semantic investigation annotation (Apply required)
investigation_replay	optional finding_id, conclusion, tools_run, evidence_times	Read-only: structured investigation replay card

Tool	Parameters / targets	Effect
what_if	change; optional task	Read-only: heuristic slice-replay what-if (migrations / blocking / load balance; not FreeRTOS kernel)
optimize_experiment	optional task, limit (1-12, default 5)	Read-only: run ranked automatic pin/priority/contention/migration experiments
analyze_traces	(none)	Read-only: rank all loaded tabs by scheduling behavior
baseline_score	optional task, baseline, snapshot	Read-only: score current per-task metrics (WCET/blocking/migrations/response) against the stored historical baseline; flags $ z > 2$
recommend_experiments	optional finding_id, task, limit (1-20, default 5)	Read-only: suggest simulation / firmware / measurement validation experiments from findings heuristics
export_investigation	optional finding_id, conclusion, tools_run, evidence_times	Download the completed investigation (finding, tools run, queries, evidence, conclusion, confidence, alternatives) as a JSON package

Tool	Parameters / targets	Effect
detect_priority_inversion	optional task, window	Read-only: scan priority-inheritance boost episodes for L/M/H inversion suspects (high/medium/low task, mutex, time, duration)
find_related_findings	optional finding_id, task, metric, window, limit (1-40, default 10)	Read-only: relate Analysis Findings by shared task, metric keyword, evidence-time proximity, or severity adjacency
compare_tasks	task_a, task_b; optional metrics	Read-only: side-by-side execution/blocking/migrations/priori inheritance delta table between two tasks
explain_finding	optional finding_id, level (quick / technical / deep)	Read-only: explain one Analysis Finding at the chosen depth (host-side; uses finding text plus hypotheses)
interpret_query	question	Read-only: turn a free-form question into an explicit investigation mode/scope before other tools run

Tool	Parameters / targets	Effect
validate_experiment	optional expected, actual (metric → signed percent)	Read-only: compare expected experiment deltas with actual A vs B / what-if results (VALIDATED / PARTIALLY VALIDATED / DISPROVED)
manage_hypotheses	hypothesis_id, status (supported / possible / rejected / need_evidence); optional reason, finding_id	Read-only: mark one investigation hypothesis status
plan_investigation	optional question, finding_id	Read-only: rank hypotheses and the cheapest tool sequence
suggest_scope	optional question	Read-only: recommend task / related tasks / time window
detect_contradictions	optional hypothesis, metrics	Read-only: verdict SUPPORTED / CONTRADICTED / INSUFFICIENT
assess_evidence_sufficiency	optional tools_run	Read-only: STOP INVESTIGATION / CONTINUE / REVISE HYPOTHESIS
cluster_findings	(none)	Read-only: group related findings into incidents

Tool	Parameters / targets	Effect
generate_fingerprint	(none)	Read-only: HIGH/MEDIUM/LOW scheduling, sync, and timing bands
find_similar_ investigations	optional limit	Read-only: match fingerprint against recorded experiment outcomes
regression_localize	optional label_a, label_b	Read-only: localize A vs B inflation to a task and region
build_causal_chain	(none)	Read-only: causal / correlated / temporal edges (never silent causation)
generate_experiment_plan	optional task, limit	Read-only: ranked firmware / what-if experiments
record_experiment_outcome	optional change, predicted, actual, quality	Read-only: store outcome for later similar-case matching
score_investigation	optional tools_run, conclusion, confidence, elapsed_ s	Read-only: evidence efficiency, cost, false-confidence, falsification, scope, stop
analyze_temporal_causality	optional task	Read-only: happens-before chain from Findings times

Tool	Parameters / targets	Effect
build_task_dependency_graph	optional task	Read-only: BTF wait/preempt/migrate/PI graph; 2-hop neighborhood + upstream tasks
decompose_response_time	optional task	Read-only: relative delay-component shares
rank_root_causes	(none)	Read-only: rank causes from findings/hypotheses
verify_claim	claim; optional claim_type, subject, object, evidence	Read-only: SUPPORTED / PARTIAL / UNSUPPORTED
challenge_conclusion	optional conclusion	Read-only: alternatives and missing evidence
investigation_memory	optional action (recall / store), record, limit	Read-only: persist/recall similar cases
cluster_incidents	optional window_ns	Read-only: time-proximity incident clusters
close_investigation	optional conclusion, confidence	Read-only: close the case envelope
analyze_distribution	optional values, metric (auto / execution / blocking / priority_inheritance / tick), task	Read-only: p50/p90/p95/p99/p99.9, stddev, CV, 3-sigma outlier rate. Statistics Query with AI... on a distribution chart harvests the open plot's samples.

Tool	Parameters / targets	Effect
analyze_periodicity	optional times, expected, source (auto / tick / sti / isr / timer / release), task, durations	Read-only: expected vs p50/p99/max, RMS and peak-to-peak jitter, kind
summarize_investigation_context	optional conclusion, tools_run	Read-only: compact investigation snapshot

Models without native tool calling can emit a fenced “btftool JSON block; the viewer renders the same GUI cards. Prefer a tool-capable model for investigation-heavy workflows.

Desktop and web behavior

BTFViewer Desktop and Web are intended to provide the **same AI investigation workflow, tool behavior, evidence model, and validation rules**. For normal use, there is no separate “Desktop workflow” or “Web workflow” to learn.

Use whichever frontend fits your environment:

- **Desktop** — convenient for local files, native save dialogs, and local/private AI endpoints.
- **Web** — convenient when you want a browser-only viewer or a hosted/development deployment.

The differences are mostly platform integration details rather than AI capabilities.

Platform detail	Desktop	Web
AI tools, Investigation Case, Evidence panel, validator	Same behavior	Same behavior
Task/event/region AI actions	Same behavior	Same behavior

Platform detail	Desktop	Web
Model picker and endpoint configuration	Supported	Supported
Reports / investigation export	Native save dialog	Browser download
In-chat diagrams	Rendered in the desktop UI	Rendered as inline browser content
Self-signed HTTPS endpoint	Can optionally allow self-signed TLS per preset	Browser/OS certificate policy still applies
Local file:// launch	Not applicable	Cross-origin requests may be blocked; use the development/preview server when needed

User takeaway: AI analysis should produce the same investigation and evidence on Desktop and Web. Platform-specific differences matter mainly when configuring endpoints, certificates, downloads, or browser networking.

Detailed platform-specific setup problems are documented in **Troubleshooting** below rather than treated as separate AI features.

Troubleshooting

Symptom	Cause	Try
Authentication	Missing key or sign-in	Settings → AI → Sign in / API key

Symptom	Cause	Try
Self-signed TLS	Private CA or self-signed HTTPS gateway	Desktop: Allow self-signed TLS
Model picker	Typed id is not served	Refresh the list and pick a served id
Web: Failed to fetch / CORS	Browser blocked a cross-origin call (file:// sends Origin: null)	Prefer npm run dev / make preview (both proxy Ollama), or see Opening the web app from file://
401 / 403	Missing or rejected key / origin	Settings → AI → Sign in or API key (OPENAI_API_KEY / GEMINI_API_KEY / OLLAMA_API_KEY; local Ollama needs none)
CERTIFICATE_VERIFY_FAILED / self-signed TLS	Private CA or self-signed HTTPS gateway	Desktop: Settings → AI → Allow self-signed TLS . Web: trust the cert in the OS/browser, use http:// on a private LAN, or use the Desktop app

Symptom	Cause	Try
Chat probe timed out / The read operation timed out	GET /models lists ids only; inference is slow or hung	Test connection POSTs /chat/completions (non-streaming, 120s). Warm the model (ollama run MODEL) and retry. Debug with the curl probe below; if curl hangs too, the gateway's chat upstream is stuck. Try "stream": true if non-stream never returns. Lower context length on a VRAM-tight local host.
Model not found	Typed id is not served	Refresh the Model list (or Test connection) and pick a served id from the dropdown, or ollama pull it
Gemini HTTP 400 thought_ signature	Gemini 3 requires a thought blob on tool follow-ups	Retry the question — the viewer echoes Gemini thought signatures

Symptom	Cause	Try
Raw “btftool JSON instead of native tool calls	Model lacks (or skips) function calling	Same cards either way (one object, JSON array, or several objects per fence) — Apply or enable Auto-apply GUI actions . Switch to qwen2.5:7b / llama3.1:8b / gpt-4o / Gemini for native calls
Ask times out (over 120s) or stays on Waiting...	Cold start, CPU offload, or VRAM spill	Stop (composer icon), warm with ollama run MODEL, retry. Use Clear between long threads. Smaller model or shorter Statistics scope if the Findings card is huge
Later turns ignore earlier facts	Chat history exceeded the context window	Clear on the AI bar, or Analysis → Query with AI... / toolbar Compare → Query with AI... for a fresh scoped prompt
Need raw AI request/response dumps (Desktop)	Debugging tool rounds / provider quirks	Settings → AI → Log MCP messages to file (off by default). Appends to ./ai_mcp_messages.log; delete when finished

Test connection curl

Same body the viewer sends for **Test connection** (replace BASE, MODEL, and KEY):

```
curl -vk --max-time 180 \  
  -H "Authorization: Bearer KEY" \  
  -H "Content-Type: application/json" \  
  -d \  
    ↪ '{"model":"MODEL","stream":false,"messages":[{"role":"user","content":"Reply  
    ↪ with JSON only: {"ok\":"true"}]}',"max_tokens':24}' \  
  BASE/chat/completions
```

Opening the web app from file://

A page opened straight from disk sends `Origin: null`, which Ollama rejects with 403 — the browser then reports only `Failed to fetch`. Serving the app over `http` avoids this entirely (`npm run dev / make preview proxy Ollama for you`), and the Desktop app is not affected at all.

To keep using `file://`, allow every origin on the Ollama side:

```
# Server started from a terminal  
OLLAMA_ORIGINS="*" ollama serve
```

```
# macOS menu-bar app (Ollama.app) – a shell variable does not reach it  
launchctl setenv OLLAMA_ORIGINS "*" # then quit Ollama and reopen it
```

Verify the change took effect; expect 200 and an `Access-Control-Allow-Origin` header:

```
curl -s -D - -o /dev/null -H "Origin: null" http://localhost:11434/v1/models  
↪ \  
  | grep -iE "^HTTP|access-control-allow-origin"
```

If a `file://` page is still refused, list the null origin explicitly with `OLLAMA_ORIGINS="*,null"`. Note that `*` lets **any** page you visit reach your local models; undo it with `launchctl unsetenv OLLAMA_ORIGINS` when done.

CLI regression gate

Desktop headless CI can compare a candidate trace to a baseline and optionally ask the configured AI for a short narrative:

Headless analyze with `--fail-on-regression`:

```
python builds/btf_viewer.py analyze candidate.btf --baseline baseline.btf
  ↳ --fail-on-regression
python builds/btf_viewer.py analyze candidate.btf --save-baseline
  ↳ /tmp/base.json
python builds/btf_viewer.py analyze candidate.btf --baseline /tmp/base.json
  ↳ --fail-on-regression --ai
python builds/btf_viewer.py ai-test --dataset tests/ai --fail-under 70
python builds/btf_viewer.py ai-test --config examples/ai/benchmark.xml -o
  ↳ AI_BENCHMARK.md
python builds/btf_viewer.py ai-test --config
  ↳ examples/ai/benchmark-selfsigned.xml --insecure
```

Or make `-C BTFViewer ai-test (AI_DATASET, AI_FAIL_UNDER)` and make `-C BTFViewer ai-test-live (AI_CONFIG, optional AI_MODELS filter, writes AI\_BENCHMARK.md)`. Dataset, scoring rules, and remaining in-app work: [Benchmark / evaluation suite](#).

See also [Export → Headless CLI](#) in the user guide.

Benchmark / evaluation suite

Offline `ai-test / runOfflineBenchmark` already ships. Live runs read **model id, base URL, TLS, and API key** from a suite XML (`--config examples/ai/benchmark.xml`) and write `AI_BENCHMARK.md`. Commands: `CLI regression gate`.

The capability matrix above is qualitative (small local vs 9B+ vs cloud). The suite turns those expectations into repeatable measurements: **which model is most reliable for BTF Viewer trace investigation**, not which model is largest or “smartest.”

Scope

Keep the live set focused on:

- **Gemini cloud models**
- **Local Ollama models that are practical on a typical developer workstation**

Do **not** pick local models only because they are newest or largest. Measure models that can run alongside BTF Viewer, Ollama, and the AI context/tooling workload.

Recommended models

Gemini (configurable; newer ids can be added without changing the runner):

- **Gemini 3.6 Flash** — high-reasoning cloud reference
- **Gemini 3.1 Flash-Lite** — fast/efficient cloud reference

Local — developer workstation:

- **Qwen3.5 9B** (`qwen3.5:9b`) — shipped in-app default; primary practical local investigator
- **Qwen3.5 27B** — higher-quality local / memory-and-latency stress test
- **Qwen3.8 27B** (`qwen3.8:27b`) — newer Qwen 27B local comparison
- **Gemma 4 26B** — non-Qwen local comparison

Older 7B/14B ids stay optional. Do not include 3B-class models — they skip native tool calls and fail the investigation suite.

Local AI – developer workstation

```
|
|— Practical / default
|   └─ Qwen3.5 9B
|
|— High-quality local
|   └─ Qwen3.5 27B
|       └─ Qwen3.8 27B
|           └─ Gemma 4 26B
```

A 9B model may beat a 27B model on this app if the larger id only slightly improves accuracy while blowing latency and memory. Measure **diagnostic quality** and **practical system performance**.

Do not hard-code the model list into the runner. Copy [examples/ai/benchmark.xml](#) (self-signed TLS: [benchmark-selfsigned.xml](#)):

```
<ai-benchmark version="1">
  <dataset>tests/ai</dataset>
  <fail-under>0</fail-under>
  <output>AI_BENCHMARK.md</output>
  <endpoint>
    <base-url>http://localhost:11434/v1</base-url>
    <tls-verify>true</tls-verify>
    <timeout-s>360</timeout-s>
  </endpoint>
  <models>
    <model id="qwen3.5:9b"/>
    <model id="qwen3.8:27b"/>
    <model id="gemini-3.6-flash" preset="gemini">
      <base-
↪ url>https://generativelanguage.googleapis.com/v1beta/openai</base-url>
      <api-key env="GEMINI_API_KEY"/>
    </model>
    <model id="gemini-3.1-flash-lite" preset="gemini">
      <base-
↪ url>https://generativelanguage.googleapis.com/v1beta/openai</base-url>
      <api-key env="GEMINI_API_KEY"/>
    </model>
  </models>
</ai-benchmark>
```

```
<!-- Self-signed / private CA gateway -->
<endpoint>
  <base-url>https://llm.internal.example:8443/v1</base-url>
  <tls-verify>false</tls-verify>
  <api-key env="GATEWAY_API_KEY"/>
</endpoint>
```

<api-key env="VAR"> reads the environment first, then any text inside the element. Omit the text (and do not commit secrets). `tls-verify false`, or `ai-test --insecure`, skips certificate checks on Desktop. `--models id1,id2` selects a subset of <model> entries. For Ollama, list the ids you actually have pulled. Record the exact model identifier and runtime configuration.

In-app picker (not shipped): **Settings** → **AI** → **Benchmark** with checkboxes for Gemini and local Ollama models, then **Run Benchmark**.

Dataset

tests/ai/ holds known traces for the major diagnostic scenarios. Each case has **expected facts**, not an exact natural-language answer:

```
tests/ai/
├─ migration_thrash.btf
├─ mutex_contention.btf
├─ priority_inversion.btf
├─ deadline_miss.btf
├─ load_imbalance.btf
├─ trace_regression.btf
├─ explain_region.btf
├─ adversarial_mutex_vs_starvation.btf
├─ adversarial_exec_vs_preemption.btf
├─ adversarial_correlation_not_cause.btf
├─ adversarial_out_of_scope_time.btf
├─ period_jitter.btf
├─ waiter_owner_handoff.btf
├─ stats_page_next_check.btf
├─ response_vs_blocking.btf
├─ preempt_matrix_vs_chain.btf
├─ mutex_block_vs_wait_queue.btf
```

Adversarial cases (kind adversarial) use a decoy finding or timestamp. The obvious answer is wrong:

Case	Decoy	Actual
adversarial_mutex_vs_starvation	mutex contention	CPU starvation / preemption
adversarial_exec_vs_preemption	long execution / WCET	preemption
adversarial_correlation_not_cause	ISR caused Comm latency	correlation, no causal link
adversarial_out_of_scope_time	diagnose jump:9000	timestamp outside the cursor window
period_jitter	tick health / tickless	task inter-arrival; open Period / Jitter
waiter_owner_handoff	kernel wait-queue	heuristic mutex handoff; open Waiter × Owner
stats_page_next_check	invent detect_timeline_anomalies	open Timeline Anomalies / Worst Events
response_vs_blocking	Blocking Time is end-to-end response	open Response Time
preempt_matrix_vs_chain	invent detect_preemption_matrix	open Preemption Matrix
mutex_block_vs_wait_queue	reconstruct the kernel wait queue	open Mutex Blocking

```
id: migration_thrash
trace: migration_thrash.btf
expected:
  finding_types: [migration, load_balance]
  tasks: [CS[22]]
  evidence:
    required_metrics: [migrations]
  allowed_tools: [detect_anomalies, correlate_events, query_raw_metric]
  forbidden:
    invented_task_names: true
    out_of_scope_timestamps: true
```

That keeps scoring robust against harmless wording differences.

Evaluation metrics

Metric	What it measures
Finding identification	Did the model identify the expected problem?
Evidence accuracy	Are cited metrics/events actually present? required_metrics also accepts Statistics page titles (Period / Jitter, Waiter × Owner, Timeline Anomalies, ...) and common aliases (Period/Jitter, Waiter × Owner)
Timestamp validity	Are jump:TIME values real and in scope?
Task-name validity	Did the model use only known task names?
Tool selection	Did it call appropriate investigation tools?
Tool-chain quality	Did it gather enough evidence before concluding?
Root-cause accuracy	Does the conclusion match the expected diagnosis?
Alternative handling	Did it consider plausible alternatives?
Confidence calibration	Is confidence consistent with the available evidence?
Response completeness	Did it answer the investigation question completely?
Latency	How long did the investigation take?
Tool-call count	How many tool rounds were required?
Peak memory	How much RAM was consumed during inference?
Time to first token (TTFT)	How quickly did the model begin responding?
Generation throughput	Sustained tokens/sec during the investigation
Investigation success rate	Percentage of cases completed correctly within the configured time/resource limit
False-causal rate	Claimed a causal link the case marks as coincidence / non-causal (0-100, higher is worse)
False-confirmation rate	Confirmed the decoy finding (trap_phrases) instead of the real cause

Metric	What it measures
Unsupported-claim rate	Share of validator claims that fail task/time/scope checks
Premature-conclusion rate	High confidence or a conclusion before required tools ran

For local runs, memory and latency are first-class. A slightly more accurate model that is unusable under memory pressure should not automatically rank higher.

Level 1 — tool / evidence correctness: valid tool, parameters, task, timestamp, and scope. Isolates tool-use bugs from reasoning quality.

Level 2 — diagnostic correctness: expected vs actual diagnosis, evidence, and alternatives. A convincing explanation is not enough.

Headline score: weighted engineering score (Finding / Evidence / Tool use / Root cause / Calibration / Safety), not a probability of correctness. Keep the component scores visible. PASS is overall ≥ 70 .

Safeguards the suite must test: no invented task names, metric values, or jump:TIME; no timestamps outside the cursor region; no unsupported conclusions presented as confirmed; evidence must match tool results; heuristic what-if stays labeled as estimates; the model must not claim a simulation is a measured result.

Model matrix

Same suite against selected Gemini and local Ollama models. Recorded 2026-08-14; full case tables: [AI_BENCHMARK.md](#).

Model	Category	Root				Notes
		Finding	Evidence	cause	Calibration	
Gemini 3.1 Flash-Lite	Cloud / fast	79	64	71	80	overall 81 , 5.5s; tool follow-up

Model	Category	Findings	Root Cause			Calibration	Notes
			Evidence	Impact	Severity		
Gemini 3.6 Flash	Cloud	—	—	—	—	—	overall 75 , 6/7; free-tier 429 split the part dump
Qwen3.5 9B	Local / practical	79	86	57	80		overall 82 , 10.7s
Qwen3.5 27B	Local / high- quality	71	71	71	80		overall 80 , 64.4s
Gemma 4 26B	Local / high- quality	71	57	86	80		overall 80 , 72.6s; 5/7 PASS

Live --config runs a tool-result follow-up when the first turn is tools-only (or planning text without a Confidence line). Single-turn scores are not comparable.

Context-size (Findings + tools + history). Do not judge a local model only by tokens/sec — check tool use and grounding as context grows:

Context	Purpose
8K	Minimum investigation workload
16K	Typical investigation
32K	Large Findings / multi-tool investigation
64K	Stress test, if supported

Practical comparison on a developer workstation:

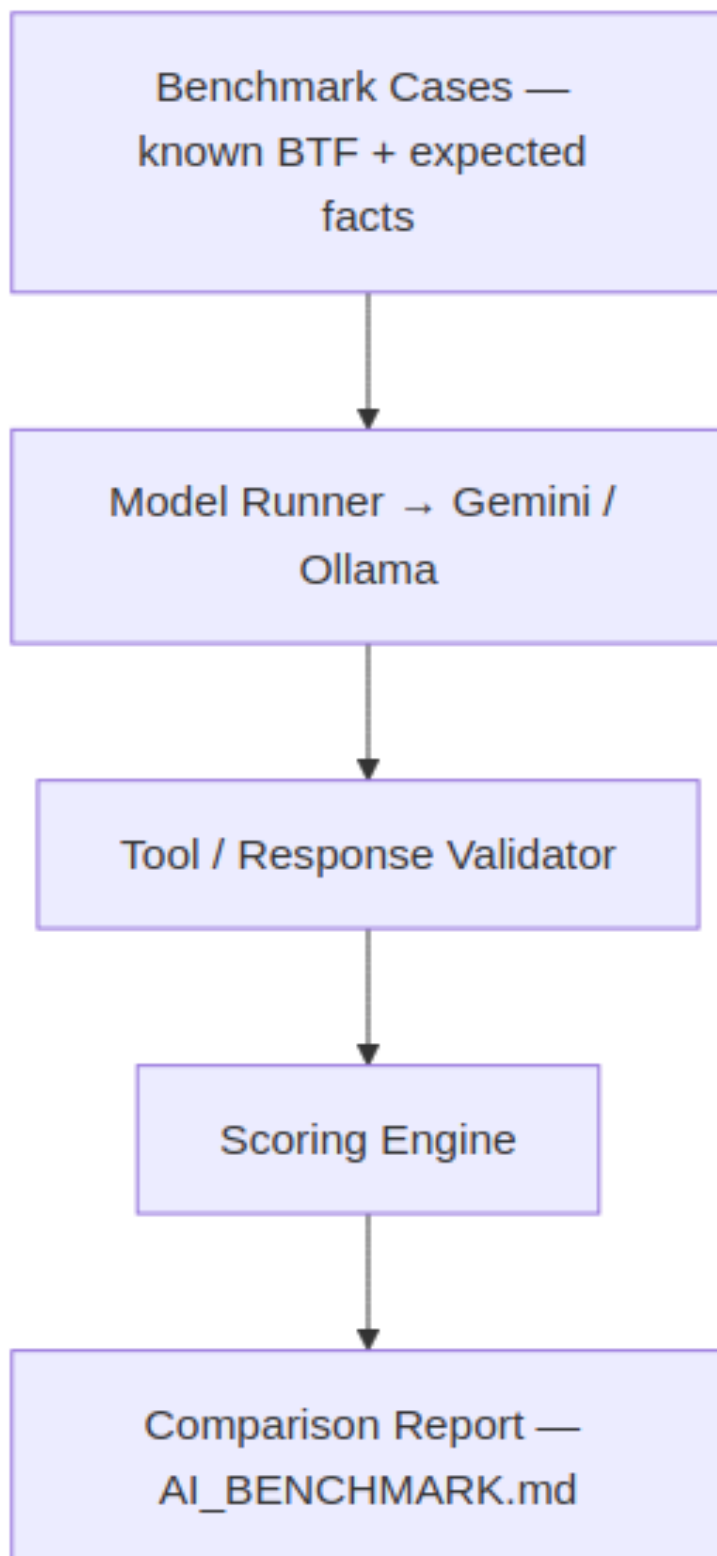
Gemini 3.6 Flash / Gemini 3.1 Flash-Lite
vs
Qwen3.5 9B (shipped default)
Qwen3.5 27B
Qwen3.8 27B
Gemma 4 26B

Does extra local capacity improve investigation quality enough to justify memory and latency?

Reproducibility and architecture

Each run should save timestamp, app version, dataset version, model ids, endpoint config, cases, prompts, tool calls/results, final responses, scores, and timing. A run ID (AI Benchmark #2026-08-13-001) keeps results comparable when model behavior drifts without a viewer code change.

--fail-under N can fail CI when a model drops below a threshold (live often wants 0 so HTTP errors still write the report).



Investigation Case

Desktop and Web share one **Investigation Case** model (btf-investigation-case): question, scope (trace / C1-Cn / tasks / cores), hypotheses with status (**supported** / **possible** / **need evidence** / **rejected**), evidence graph, coverage, falsification checks, conclusion, and validation. Lockstep notes: **Implementation notes**.

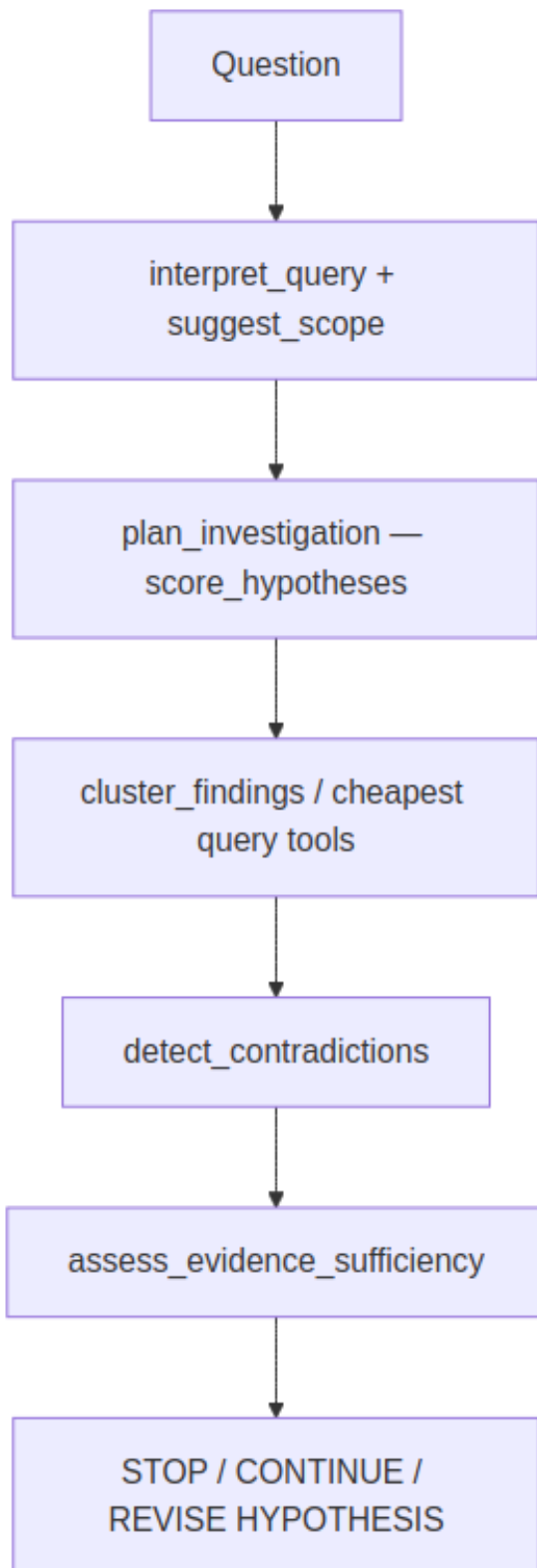
After each final assistant reply a host-side **validator** extracts jump:TIME and Task[id] claims and flags invented names or timestamps outside the cursor window. **Test connection** appends a **model capability** card (live chat / structured output / tool calling overlay on a 3B vs 7B+ heuristic). Headless eval:

```
make -C BTFViewer ai-test  
# or: python builds/btf_viewer.py ai-test --dataset tests/ai --fail-under 70
```

Modes (Quick / Diagnose / Compare / Optimize / Report) map onto existing templates; they do not add new tools. Always confirm on the timeline.

Investigation planner

Host-side planner. **Cheapest evidence first**. User-facing loop: **README → Investigation planner**.

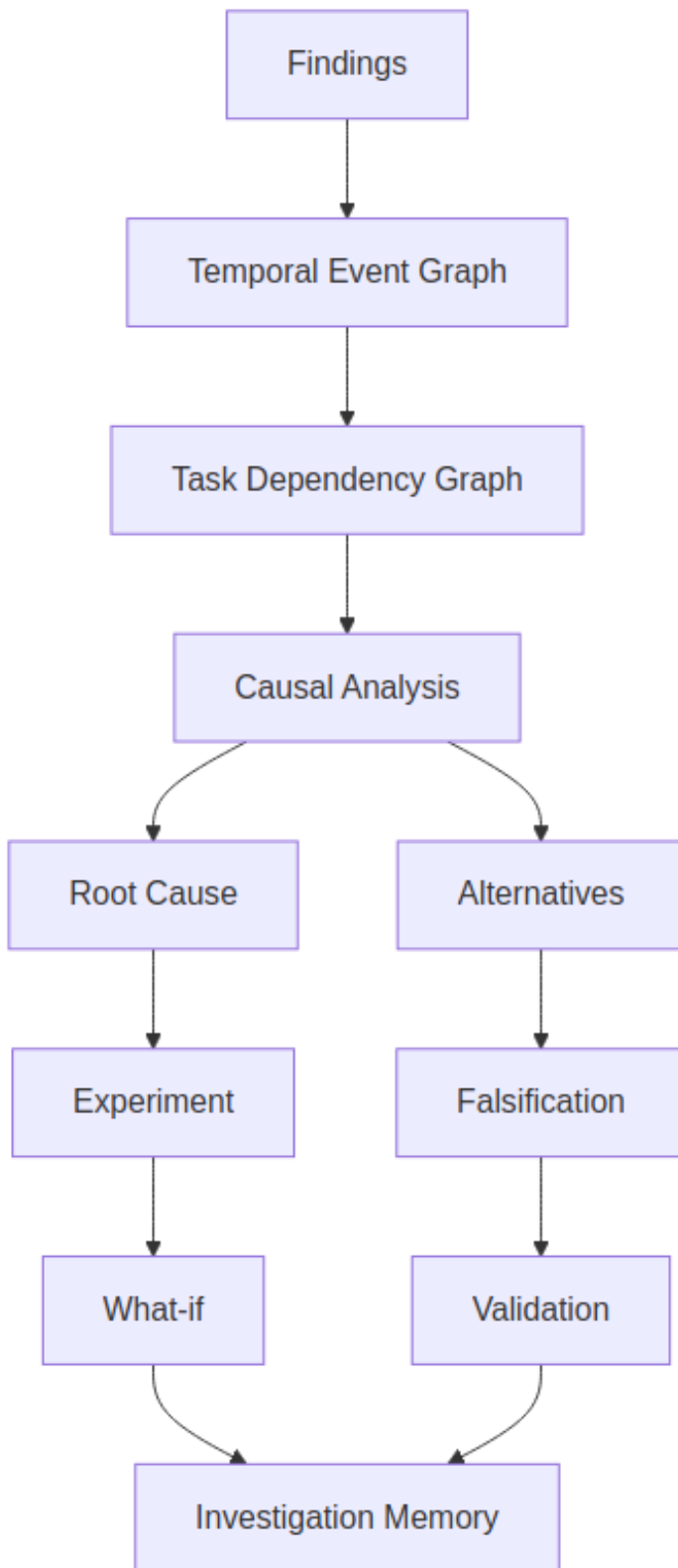


Tool / helper	Host behaviour
<code>plan_investigation</code>	Rank hypotheses and a cheap tool sequence from findings + question
<code>suggest_scope</code>	Task, related tasks, evidence times (or current cursors)
<code>detect_contradictions</code>	SUPPORTED / CONTRADICTED / INSUFFICIENT (e.g. execution $\gg$ blocking vs mutex hypothesis)
<code>assess_evidence_sufficiency</code>	Coverage heuristic $\rightarrow$ stop / continue / revise
<code>score_hypotheses</code>	Evidence-weighted scores (not a GUI tool)
<code>cluster_findings</code>	Group by shared task or pattern
<code>generate_fingerprint</code>	HIGH / MEDIUM / LOW scheduling, sync, timing bands
<code>find_similar_investigations</code>	Jaccard-style match vs recorded experiment outcomes
<code>regression_localize</code>	A vs B deltas $\rightarrow$ task / region / likely mechanism
<code>build_causal_chain</code>	Edges tagged causal / correlated / temporal; disclaimer required
<code>generate_experiment_plan</code>	Ranked pin / contention / priority experiments
<code>record_experiment_outcome</code>	Persist outcome (Desktop [ai] experiment_outcomes, Web localStorage)
<code>score_investigation</code>	Phase 3 extras: evidence_efficiency, investigation_cost, false_confidence, falsification_quality, scope_accuracy, stop_efficiency (also spread into score_benchmark_case, with adversarial rates)

Do **not** add chat templates after `auto_investigate`.

Causal and temporal engines

Host-side heuristics over Analysis Findings — not a FreeRTOS scheduler replay. User-facing loop stays on **Investigation planner**. Diagnose / Investigate / Auto investigate walk explanation tools before experiments: graph → temporal → rank_root_causes → challenge_conclusion, then what_if.



Tool / helper	Host behaviour
<code>analyze_temporal_causality</code>	Happens-before chain from finding times (jump:TIME)
<code>build_task_dependency_graph</code>	BTF sync/preempt/migrate/PI graph (finding-wording fallback); optional task neighborhood
<code>decompose_response_time</code>	Relative delay shares (mutex, preemption, migration, execution, scheduler)
<code>rank_root_causes</code>	Rank hypotheses or finding buckets
<code>verify_claim</code>	SUPPORTED / PARTIAL / UNSUPPORTED vs findings and cursors
<code>challenge_conclusion</code>	Alternatives and missing evidence
<code>investigation_memory</code>	Store/recall (Desktop [ai] investigation_memory, Web localStorage)
<code>cluster_incidents</code>	Group findings by time proximity
<code>close_investigation</code>	Record conclusion and close the case
<code>analyze_distribution</code>	p50 / p90 / p95 / p99 / p99.9, stddev, CV, 3-sigma outlier rate; BTF execution/blocking/PI/tick harvest
<code>analyze_periodicity</code>	Period/jitter from tick, STI, ISR, timer, or task-release times; kind = drift vs jitter vs WCET vs scheduler
<code>summarize_investigation_context</code>	Compact findings, hypotheses, and tools run

Engine limits

Engine	What it is	What it is not
analyze_temporal_causality	Happens-before from finding jump:TIME	Kernel event replay
build_task_dependency_graph	BTF sync / preempt / migrate / PI edges; 2-hop task neighborhood	Full ISR / object graph
decompose_response_time	Relative shares from finding magnitudes	Cycle-accurate milliseconds
rank_root_causes	Hypothesis or finding-bucket rank	A probability
investigation_memory	Local store / recall notepad	Team knowledge base
cluster_incidents	Time-proximity groups	Shared-mutex / causal clustering
close_investigation	Case status closed plus conclusion	Full firmware A/B lifecycle
analyze_distribution	BTF execution / blocking / PI / tick samples (cap 8000)	A response-time series the parser does not have
analyze_periodicity	Inter-arrival jitter and kind	A kernel period timer
simulate_schedule	LEVEL 1 helper inside what_if	A GUI tool or FreeRTOS kernel

Out of scope (do **not** add chat templates for these): trace-to-code (ELF / DWARF), real scheduler or hardware-aware simulation, model routing, automatic benchmark-case generation, natural-language → metric compiler, anomaly discovery without Analysis Findings, shared team investigation database.

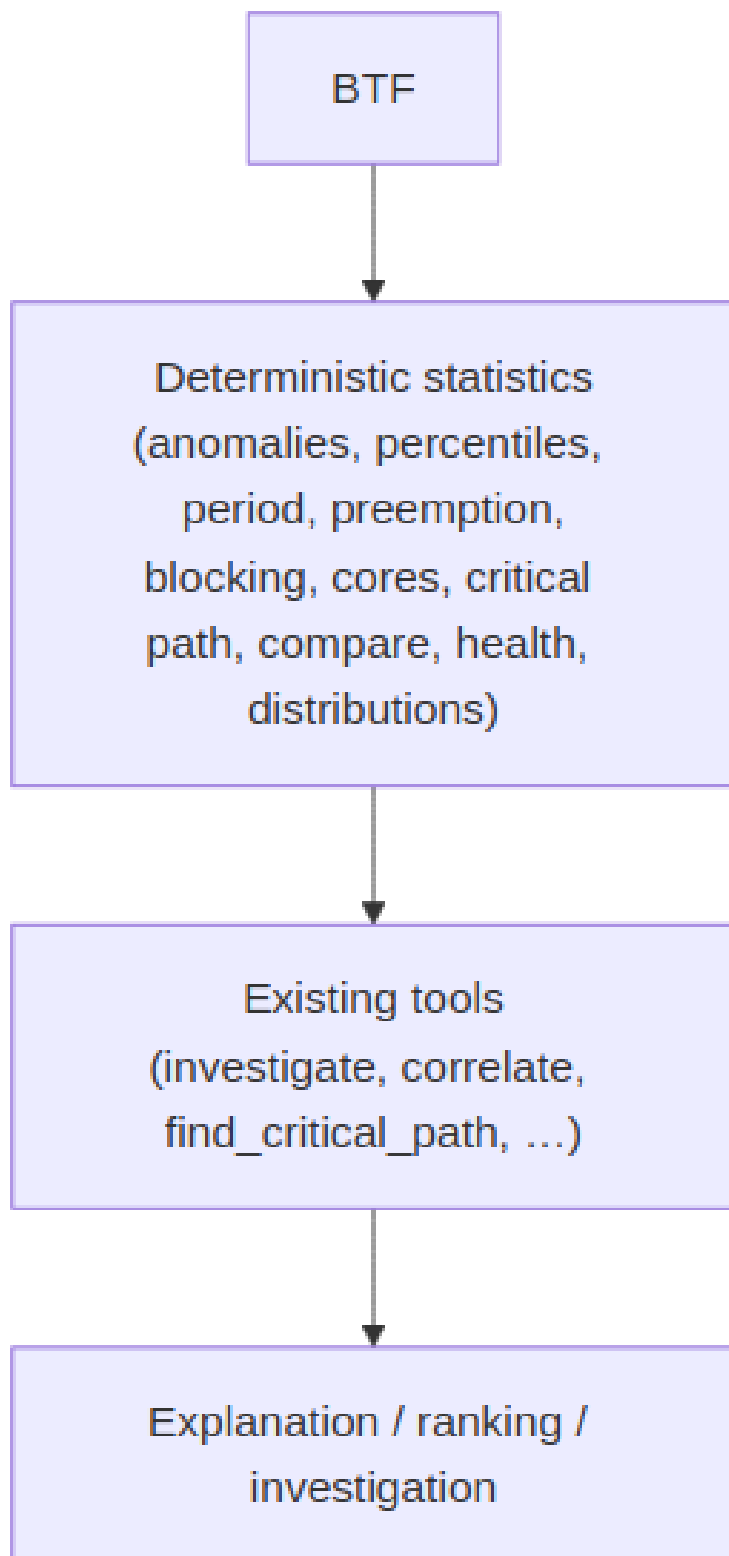
Do **not** add chat templates after auto_investigate. Next gains come from deeper engines, not more buttons.

Implementation notes

Tech notes for keeping Desktop and Web in lockstep. User-facing Case / Evidence behaviour: [README → Investigation Case](#). Live suite XML: [Benchmark / evaluation suite](#). Recorded scores: [AI_BENCHMARK.md] (AI_BENCHMARK.md).

Analysis vs AI tools

Facts come from BTF Statistics pages first. AI ranks, explains, and navigates those facts. **Do not add AI tools** for work those pages already do (no detect_timeline_anomalies, extra jitter tools, or histogram tools). **Do not** invent kernel response time, inspect ELF/source, or simulate the scheduler.



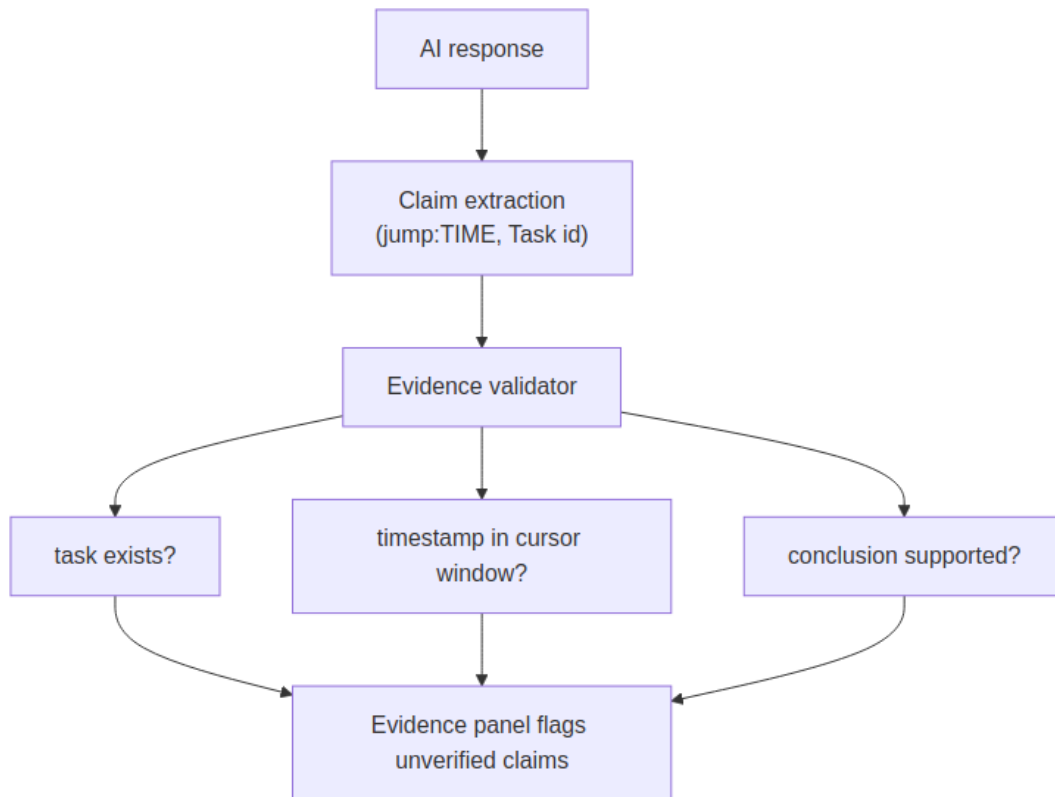
The shipped loop stays **Triage → Investigate → Verify → Correlate → Critical Path → Dependency Graph → Temporal Causality → Rank → Challenge → What-if → Report**. Improve access to Statistics evidence; do not grow the tool list. User-facing page map: [README](#) → [BTF analysis pages](#).

Shared Case / Evidence engines

Desktop and Web share one Case, Evidence, planner, causal, tool, and mermaid implementation. After AI UI changes: make `-C BTFViewer bundle` and make `-C BTFViewer web`.

UI lockstep: mode chips wrap. Primary templates are two rows: Analysis Findings / Explain region / Investigate, then **Auto investigate + More templates....** Chip min-height 28px, disabled chips/menu items #8a96a8. Findings **Investigate...** uses the same outline style as the other Analysis footer buttons (not accent/primary). **More** templates use the same groups in a 2-column overlay. Trace Compare opens from toolbar **Compare**, not the Statistics footer.

The Desktop ai-test CLI and the Web offline benchmark share tests/ai fixtures (tracked `.btf stubs + dataset.json`).

Validator

The host validator runs after the final reply. Prompting still forbids inventing numbers, task names, and `jump:TIME`; the validator is the guard, not the prompt.

Experiment close-out

`validate_experiment` compares expected vs actual signed percents (VALIDATED / PARTIALLY VALIDATED / DISPROVED), updates open hypotheses, and offers **Save to knowledge** (`btfexp:save`). Empty actual is filled from the last Trace Compare refresh (including **Scope to cursors**) or `compare_performance` via `experiment_percents_from_compare`. Toolbar **Compare** → **Validate experiment...** closes the dialog and asks the model to call the tool with actuals omitted. Firmware-change capture remains a user step (new trace + toolbar **Compare**).

Capability, cost, privacy, knowledge

Feature	Host behaviour
Capability probe	Test connection lists models, chats with a JSON structured-output probe, then tool-calling (btf_ping then btf_pong). Live results overlay chat / structured output / tool calling / multi-tool chaining; long context and reasoning stay heuristic.
Cost	A dedicated usage bar shows accumulated tok · tools · s (and estimated USD when priced). Evidence uses the full format_cost_meter line. Clear resets replies, the meter, and current investigation issues.
Privacy	Chip ● Local / ● Cloud / ● Sensitive. Cloud send is blocked when sensitive; otherwise annotations are sanitized and optional task-name aliases apply (apply_cloud_privacy).
Knowledge	investigate matches user-saved entries (More → Save current finding...), then baseline, then the builtin catalog. Typical vs current rates show when both exist.
Interpret	Free-form Ask host-interprets first (interpret_query). Templates / modes / Run investigation skip confirm.
Tool Why?	Evidence Investigation lists each tool with a host-side reason (btftool:why/name).

Diagrams

Replies may include “mermaid **sequence** diagrams (mutex take/give, block/resume, priority boost / L/M/H) and graph LR / **flowchart** core-migration graphs (counts on edges). Pipe **Markdown tables** (and a sanitized HTML <table> copied from Findings) render as HTML tables in the reply pane. In-chat Markdown / HTML tables use the same sanitizer. The Evidence panel adds its own auto-generated graph

TD **Investigation tree** (root-cause chain steps as boxes, alternative hypotheses as rounded nodes branching off the finding) whenever investigate returns a chain — same renderer, same click/zoom rules below. Sequence, flowchart, and investigation-tree palettes follow **dark and light** theme; **Save As...** HTML conversation export uses the **light** palette for white paper.

- Click a **task** node to lock-highlight that timeline row (Low[266] (Core 0) resolves to Low[266]).
- Click a **core** node (Core_0, C0, C1) to switch to Core View and scroll to that core.
- Mutex hex and other unresolved labels do nothing (the timeline stays undimmed).
- Click empty figure area to open a larger zoom window (scroll to zoom 0.5–6×; **Esc** or **Close**). Trackpad pinch is treated as scroll.
- The link row under the figure has the same targets.
- **Save As...** HTML keeps inline SVG with clickable nodes (chat zoom wrappers are omitted).

Documentation navigation

Document	Question answered
README.md	How do I use BTFViewer?
WORKFLOWS.md	How do I diagnose a problem?
STATISTICS.md	What does this measurement mean?
AI.md	How does AI-assisted investigation work?
